

Lesson 09 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 20 items are *[Original]* JKSSB-pattern questions for this lesson. No item is a real PYQ, and the answer key is the author's verification of the lesson's concepts.

- Q1. B.** Virtual memory uses secondary storage to extend the addressable memory beyond physical RAM. A wrongly calls it extra RAM; C describes cache; D describes ROM.
- Q2. C.** Virtual memory is slower than RAM because it is disk-backed. A and B overstate its speed; D is false, since virtual memory is normally in use.
- Q3. B.** Virtual memory is disk-backed and slower than RAM, extending capacity rather than adding RAM. A contradicts this; C is false because cache is faster; D is false because RAM is still required.
- Q4. B.** Virtual memory lets programs and data larger than physical RAM run and gives each process a large contiguous view. A, C, and D describe unrelated hardware changes.
- Q5. A.** MMU stands for Memory Management Unit. B, C, and D are invented expansions.
- Q6. A.** The MMU translates virtual addresses into physical addresses at run time. B is the job of secondary storage; C and D describe other components.
- Q7. B.** In paging, the fixed-size virtual blocks are pages. Frames (C) are the physical blocks; segments (A) belong to segmentation; sectors (D) are disk units.
- Q8. B.** The fixed-size physical blocks into which pages are loaded are frames. Pages (A) are the virtual blocks; segments (C) belong to segmentation; registers (D) are CPU storage.
- Q9. B.** Paging uses fixed-size pages and frames, while segmentation uses variable-size logical segments. A and D reverse the two; C wrongly makes both fixed-size.
- Q10. B.** Segmentation divides a program into variable-size logical segments. A, C, and D describe paging or cache rather than segmentation.
- Q11. B.** A page fault means the referenced page is not currently in RAM and must be loaded from disk. A calls it a hardware fault, which is wrong; C and D describe unrelated failures.
- Q12. A.** Demand paging loads a page only when it is referenced and not already present. B describes loading everything at once; C and D are false.
- Q13. C.** Virtual memory is *not* additional physical RAM; it is disk-backed and slower than RAM. A, B, and D are correct statements, so C is the one that is NOT correct.

- Q14. A.** Statements I and II are correct; III is false because virtual memory is not extra physical RAM but disk-backed storage. B omits II; C includes false III and omits I; D includes false III.
- Q15. A.** Swapping moves data or pages between RAM and secondary storage. B, C, and D describe unrelated operations.
- Q16. A.** Thrashing is excessive paging that leaves little time for useful work. B, C, and D misdescribe it.
- Q17. B.** Cache is small, fast memory close to the CPU, whereas virtual memory is slower and disk-backed. A reverses the two; C wrongly makes both secondary storage; D swaps their purposes.
- Q18. B.** When the working set exceeds RAM, the OS uses virtual memory, keeping some pages on disk and loading them on demand. A is false; C is impossible (disk is not converted into RAM chips); D does not address memory pressure.
- Q19. C.** Disk-backed virtual memory is the slowest of these. L1 cache (A) and a CPU register (D) are fastest, and RAM (B) is faster than the disk.
- Q20. A.** Paging uses fixed-size pages and frames. B and C mismatch the block rule; D wrongly assigns file storage to the MMU.
- Q21. A.** The page/swap file on disk backs virtual memory. B, C, and D are peripherals, not memory backing.
- Q22. A.** The page file resides on disk storage. B, C, and D are not file stores.
- Q23. A.** Thrashing is pathological paging overhead. B, C, and D describe unrelated situations.
- Q24. A, B, C.** Paging, the swap file, and disk backing all belong to virtual memory. The spooler (D) serves printing.
- Q25. A, B, C.** Fixed-size paging, page-fault loading, and the thrashing definition are correct. D is false: virtual memory is a technique, not a chip.
- Q26. A.** Virtual memory uses storage space, usually on an SSD/HDD, as backing for memory pages. Registers, printer buffers, and firmware ROM do not provide this backing store.
- Q27. A.** A page fault signals that the referenced page is not resident in RAM and must be fetched. A cache hit, power failure, and print spool are different events.
- Q28. A.** Virtual addressing can provide a larger apparent address space than physical RAM. Registers and cache do not define the process's full virtual address space.
- Q29. A.** Virtual memory uses storage as backing for memory pages, while cache keeps copies of data for speed. Neither is a permanent backup, OS replacement, or register.
- Q30. A, B, C.** The OS manages virtual memory, faults locate missing pages, and backing storage can hold non-resident pages. Storage remains much slower than registers.